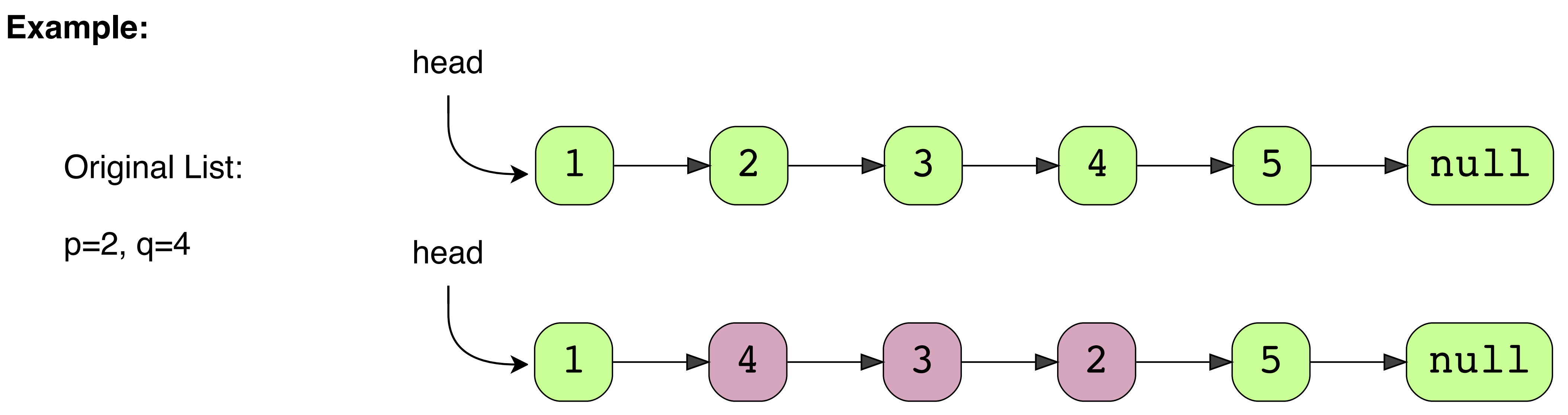
Solution: Reverse a Sub-list

Problem Statement

Given the head of a LinkedList and two positions ‘p’ and ‘q’, reverse the LinkedList from position ‘p’ to ‘q’.



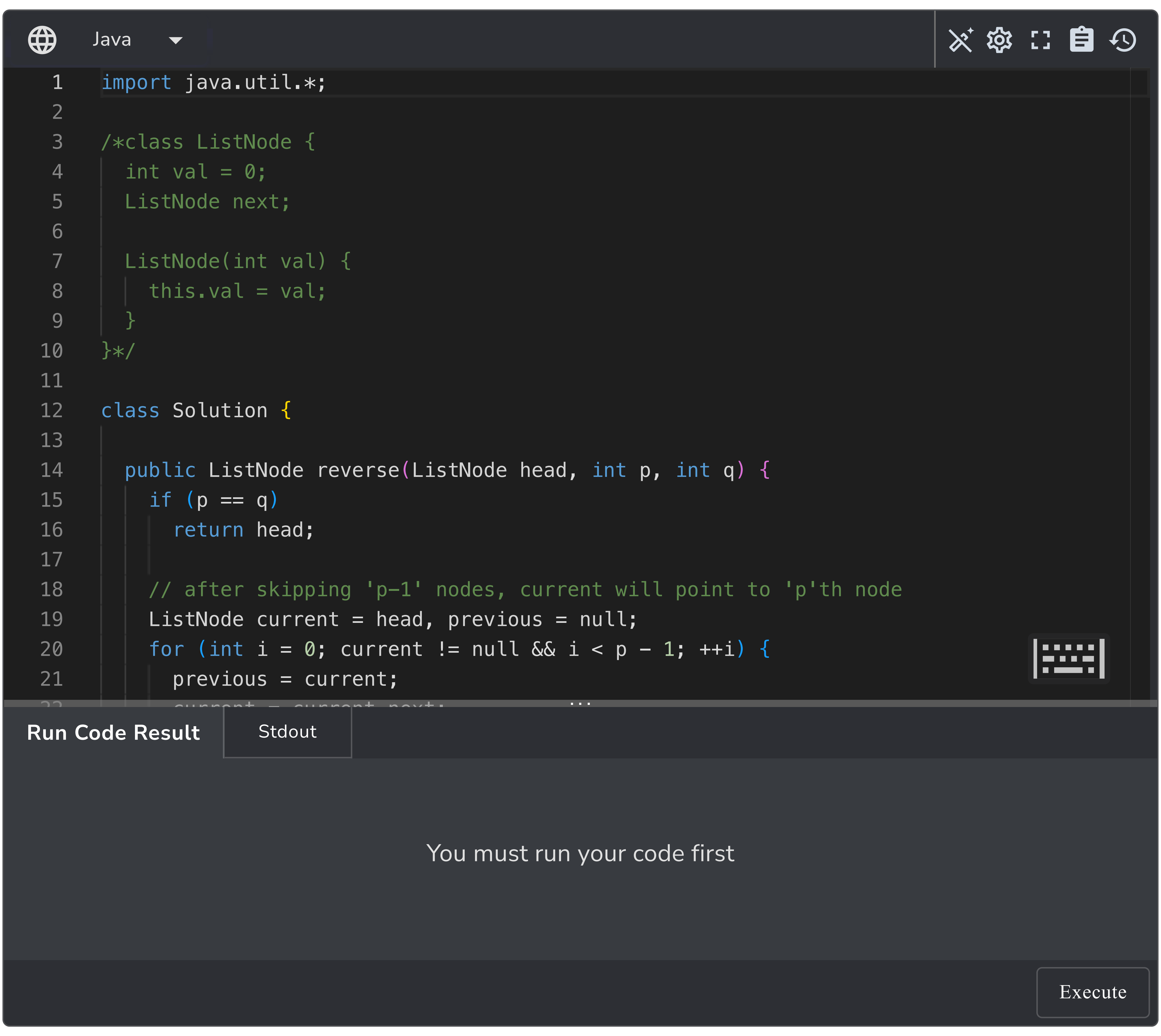
Solution

The problem follows the **In-place Reversal of a LinkedList** pattern. We can use a similar approach as discussed in `Reverse a LinkedList`. Here are the steps we need to follow:

1. Skip the first `p-1` nodes, to reach the node at position p.
2. Remember the node at position `p-1` to be used later to connect with the reversed sub-list.
3. Next, reverse the nodes from `p` to `q` using the same approach discussed in Reverse a LinkedList.
4. Connect the `p-1` and `q+1` nodes to the reversed sub-list.

Code

Here is what our algorithm will look like:



```java
import java.util.*;

/*class ListNode {
  int val = 0;
  ListNode next;

  ListNode(int val) {
    this.val = val;
  }
}*/

class Solution {

  public ListNode reverse(ListNode head, int p, int q) {
    if (p == q)
      return head;

    // after skipping 'p-1' nodes, current will point to 'p'th node
    ListNode current = head, previous = null;
    for (int i = 0; current != null && i < p - 1; ++i) {
      previous = current;
```

You must run your code first

Time Complexity

The time complexity of our algorithm will be O(N) where ‘N’ is the total number of nodes in the LinkedList.

Space Complexity

We only used constant space, therefore, the space complexity of our algorithm is O(1).

Similar Questions

Problem 1: Reverse the first ‘k’ elements of a given LinkedList.

Solution: This problem can be easily converted to our parent problem; to reverse the first ‘k’ nodes of the list, we need to pass `p=1` and `q=k`.

Problem 2: Given a LinkedList with ‘n’ nodes, reverse it based on its size in the following way:

1. If ‘n’ is even, reverse the list in a group of n/2 nodes.
2. If n is odd, keep the middle node as it is, reverse the first ‘n/2’ nodes and reverse the last ‘n/2’ nodes.

Solution: When ‘n’ is even we can perform the following steps:

1. Reverse first ‘n/2’ nodes: `head = reverse(head, 1, n/2)`
2. Reverse last ‘n/2’ nodes: `head = reverse(head, n/2 + 1, n)`

When ‘n’ is odd, our algorithm will look like:

1. `head = reverse(head, 1, n/2)`
2. `head = reverse(head, n/2 + 2, n)`

Please note the function call in the second step. We’re skipping two elements as we will be skipping the middle element.

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Reverse A Sub-List (Medium)

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Reverse Every K-Element Sub-List (Medium)

☐ Mark as Completed